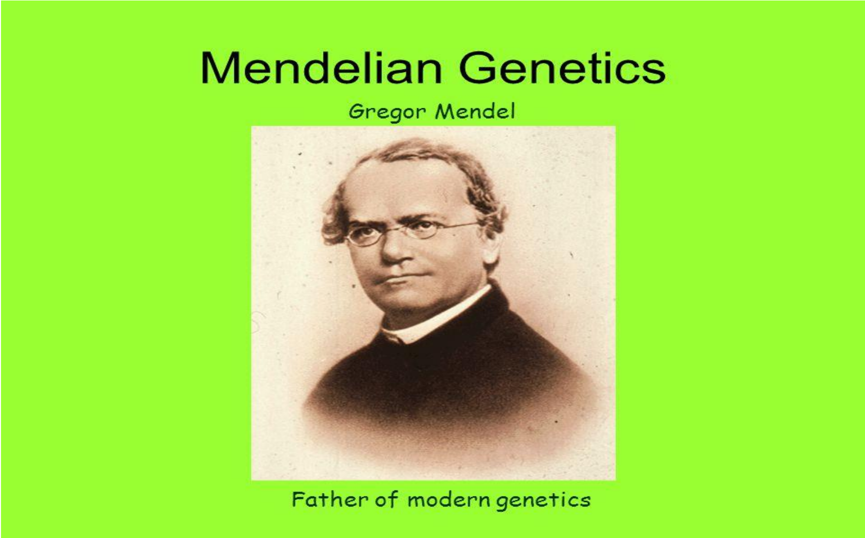
MENDELIAN GENETICS (History)

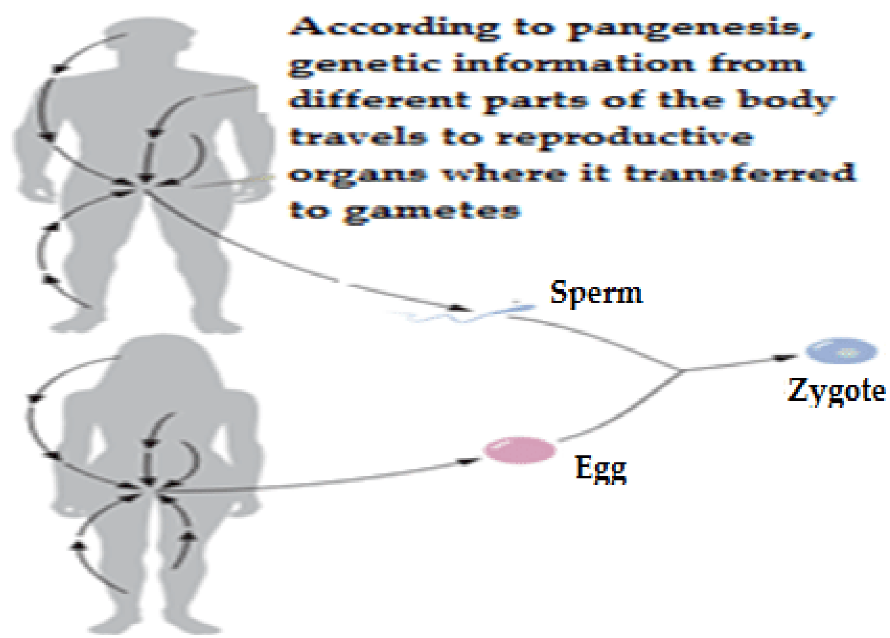


-The first quantitative studies of inheritance were carried out by an Austrian monk, **Gregor Johann Mendel** (1822-1884). He is widely considered a pioneer in the field of modern genetics. Before Mendel began his experiments, various theories about heredity had been proposed.

Historical theories about heredity

1-Pangenesis :

- The specific particles later called gemmules, carry information from various parts of the body to the reproductive organ from where they are passed to the embryo at the moment of conception. Although incorrect, this concept was highly influential and persisted until the late 1800s.

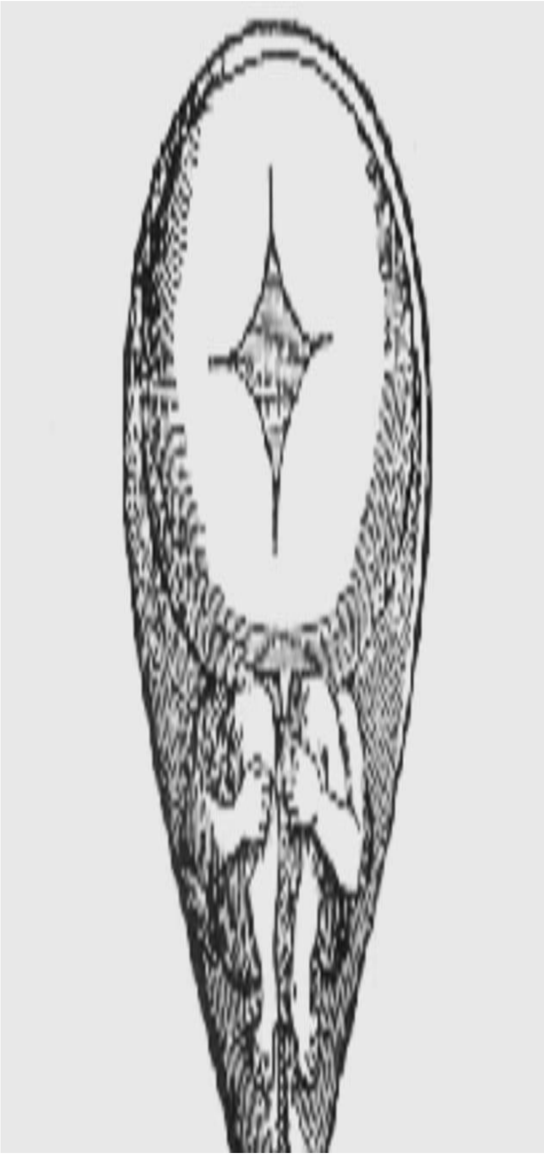


2- Preformationism:

-Inside the egg or sperm exists a tiny miniature adult, a *homunculus*, which simply enlarged during development. -Ovists argued that the *homunculus* resides in the egg, whereas spermists insisted that it is in the sperm.

-Preformationism meant that all traits would be inherited from only one parent.

- Although incorrect, this concept was remained popular throughout 17th and 18th centuries.



3-Inheritance of acquired characteristics:

- According to this notion the traits acquired during one's lifetime become incorporated into one's hereditary information and are passed on to offspring.

-For example, a person who developed musical ability through his life would produce children who are innately endowed with musical ability.

-This notion also is no longer accepted, but it remained popular through the 20^{th} century.

4- **blending theory of inheritance:**

-the idea which believed that both sexes contribute equally to a new individual. They felt that parents of contrasting appearance always produce offspring of intermediate appearance(offspring are a blend or mixture of parental traits).

-For example, across between plants with red flowers and plants with white flowers, would always produce plants with pink flowers, but the red and white flowers reappeared in future generation.

- Once blended the genetic differences could not be separated out just as green paint which cannot be separated into blue and yellow pigments. We realize today that individual genes do not blend.

5- Germ-plasm theory(chromosomal theory):

The cells of reproductive organ carry a complete set of genetic information that is transferred directly to the gametes later to offspring.

Mendel discovered the basic principles of heredity by breeding garden peas in carefully planned experiments. This work led to a paper, published in (1866) in which he demonstrated that heredity is governed by a few simple rule and that rather than being complex and mysterious. Thus the laws of heredity must explain not only the stability but also the variation that is observed between generations.

Mendel formulated two fundamental laws of heredity

Mendel's first law (the principle of segregation):

Every individual carries pair of factors for each trait and the members of the pair of alternative characters are separated during the formation of gametes.

Mendel's second law (the principle of independent assortment):

When gametes are formed, the alleles of agene for one trait segregate independently from the alleles of agene for another trait.

First law

P TT X tt

G T t

F1 Tt

100%tall

Tt X Tt

T t T t

F2 TT Tt Tt tt

Phenotypic ratio 3:1

75% dominant 25% recessive

tall short

T: allele of tall trait.

t: allele of short trait.

G > g T>t

Second law

TTGG X ttgg

TG tg

TtGg

100% tall green plant dihybrid

TtGg X TtGg

	TG	Tg	tG	tg
TG	TTGG	TTGg	TtGG	TtGg
Tg	TTGg	TTgg	TtGg	Ttgg
tG	TtGG	TtGg	ttGG	ttGg
tg	TtGg	Ttgg	ttGg	ttgg

9:3:3:1

G: allele of green trait.

g: allele of yellow trait.